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Learn from a moth: building a pavilion with no mould

The free guide to the reel on the ICD/ITKE Research Pavilion 2016-17: how a moth larva's silk inspired a robotic fibre pavilion, and how to try the same idea with string.



How it works

Fibre shells usually need a big mould, and the mould is thrown away. Some leaf miner moth larvae spin silk in a different way: they stretch a hammock between points on a bent leaf, with no mould at all.

The research team at the University of Stuttgart (ICD and ITKE), with biologists from Tübingen, copied this. Robots wound glass and carbon fibre between points on a frame, so the fibre builds the shape itself. The frame is bent into shape, and the fibre stiffens it.

The span was longer than one robot can reach, so several machines handed the fibre to each other. Press accounts describe a custom drone doing the passing.

Key facts

The pavilion is one cantilever about 12 m long, with a surface of around 40 m² (official page; sources disagree on its weight). The method was later developed into the BUGA Fibre Pavilion in Heilbronn (2019), which spans 23 m with 60 carbon and glass fibre components and a prestressed ETFE membrane.

Built examples

ICD/ITKE Research Pavilion 2016-17, Stuttgart	Bending-active frame plus coreless wound fibre, inspired by leaf miner moth larvae.
BUGA Fibre Pavilion, Heilbronn (2019)	60 robotically wound components with no mould; a single component took up to 250 kN in tests.

Try it yourself

1. Make a frame from two cardboard rings or two sets of nails in a wooden board, and fix it on the table.
2. Tie a long piece of string to one point and wind it between the points in different directions.
3. Brush the string with diluted PVA glue or flour paste, and let it dry.
4. Remove the temporary frame if you used one, and look at the shape the string has made.

5 Change the points and the winding order, and compare how stiff the results are.

6 For a digital version, draw the winding paths in Grasshopper (Rhino) as straight lines between points.

The rule to remember

Before you design a shell, ask how nature builds it.

Copying the form only	Copy the process: how it is made, not just what it looks like.
Ignoring the material	Fibre only works when it is loaded along its direction.
Forgetting scale	A tiny silk hammock does not scale directly to a building.

Go further

ICD/ITKE Research Pavilion 2016-17, University of Stuttgart
https://www.icd.uni-stuttgart.de/projects/icditke-research-pavilion-2016-17/

Larvae silk-inspired pavilion, Architect Magazine
https://www.architectmagazine.com/?p=348208

BUGA Fibre Pavilion, Dezeen
https://www.dezeen.com/2019/05/08/university-stuttgart-biomimetic-pavilion-bundesgartenschau-horticultural-show/

Credits
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Facts: ICD/ITKE Stuttgart, Architect Magazine, Dezeen, Designboom.

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